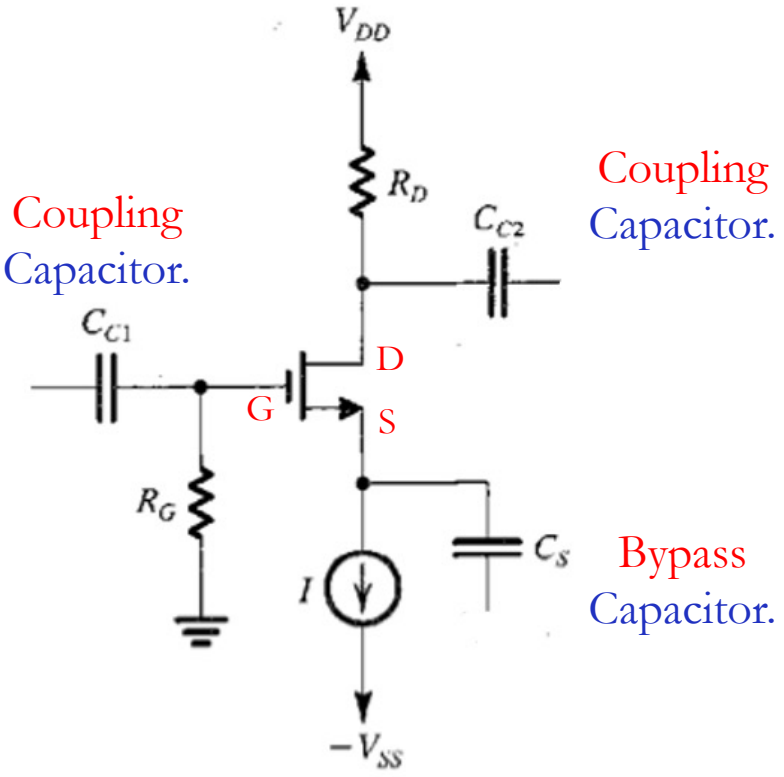
Field Effect Transistors

- MOSFET Basic Amplifier Configurations

MOSFET Basic Amplifier Configurations

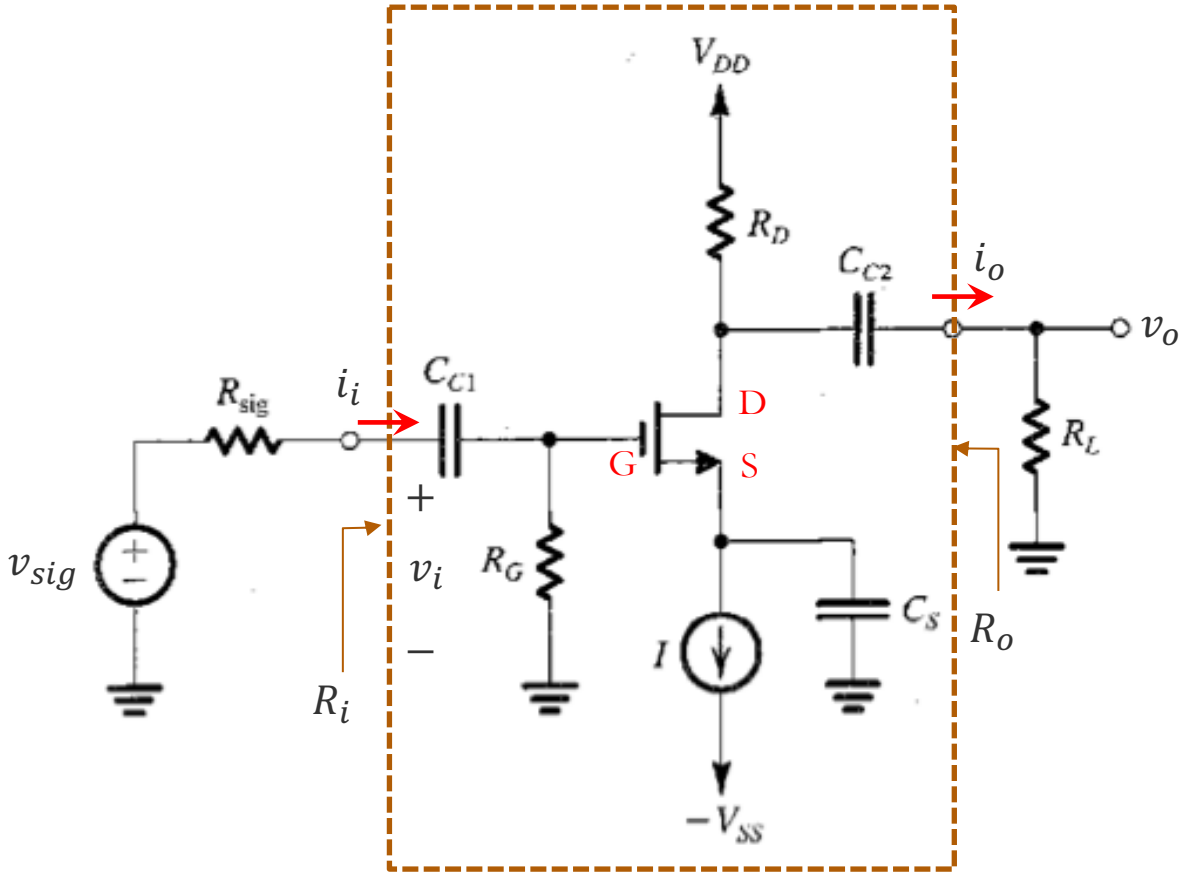
Different combinations of input-output connections are possible; each gives specific amplifier properties.

Input	Output	Ground	Name
Gate	Drain	Source	Common-source (CS)
Gate	Source	Drain	Common-drain (CD)
Source	Drain	Gate	Common-gate (CG)



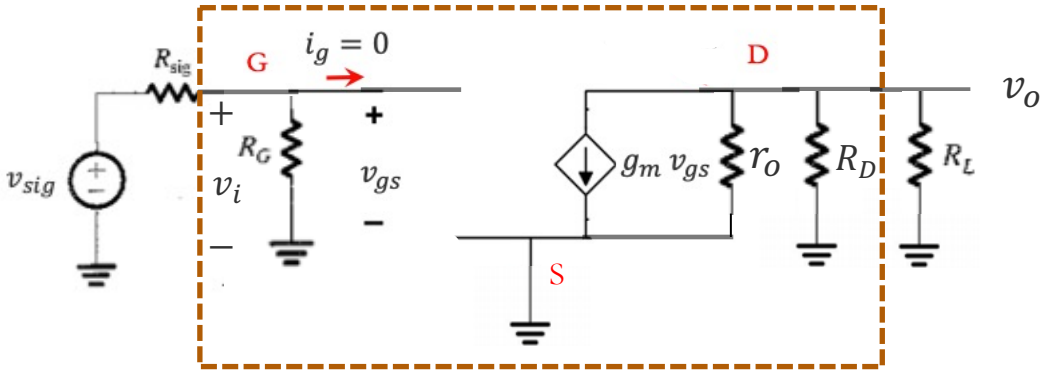
The Common-source (CS) Amplifier

This configuration works as "all-purpose amplifier": high voltage and current gains.



The Common-source (CS) Amplifier

This configuration works as "all-purpose amplifier": high voltage and current gains.



$$v_i = v_{gs} \qquad v_i = v_{sig}\frac{R_G}{R_{sig}+R_G}$$

$$v_o = -g_m v_{gs}\,(r_o \parallel R_D \parallel R_L)$$

$$v_o = -g_m v_i\,(r_o \parallel R_D \parallel R_L)$$

$$A_v = \frac{v_o}{v_i} = -g_m\,(r_o \parallel R_D \parallel R_L)$$

Voltage gain

$$A_v = \frac{v_o}{v_{sig}} = -g_m \frac{R_G}{R_{sig}+R_G}\,(r_o \parallel R_D \parallel R_L)$$

Overall Voltage gain

Open-circuit Voltage gain (no load)

$$A_{vo} = \frac{v_o}{v_i} = -g_m\,(r_o \parallel R_D)$$

Input Resistance

$$R_i = R_G$$

Output Resistance
short-circuit source v_{sig}

$$R_o = r_o \parallel R_D$$

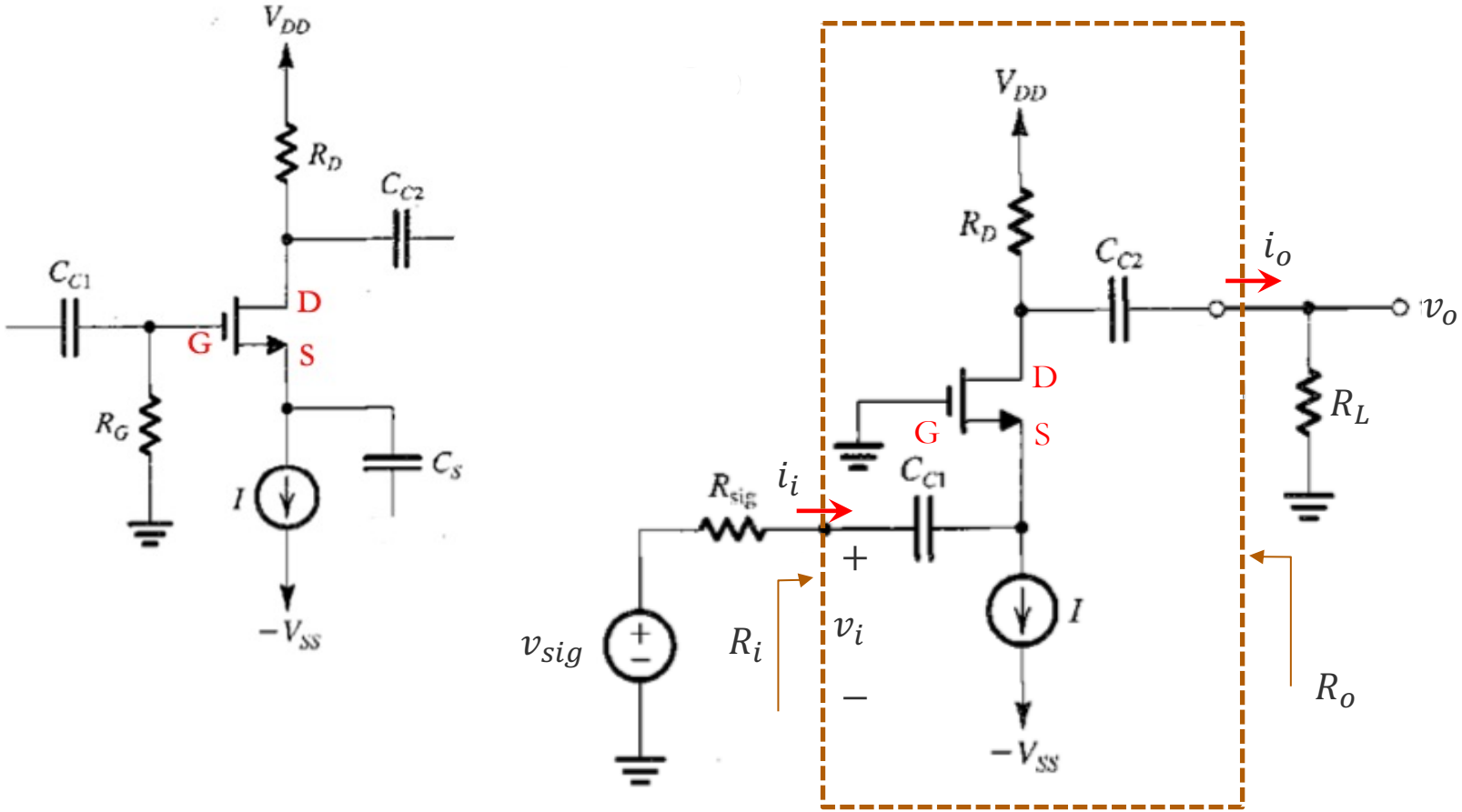
Current gain

$$A_i = \frac{i_o}{i_i} = \frac{v_o}{R_L}\cdot\frac{R_i}{v_i}$$

$$A_i = \frac{i_o}{i_i} = A_v \cdot \frac{R_i}{R_L}$$

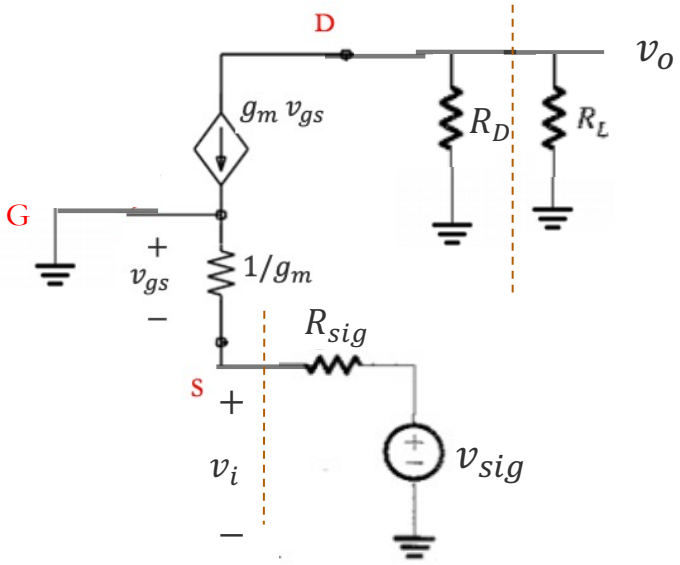
The Common-gate (CG) Amplifier

This configuration works as a "current buffer": unity current gain.



The Common-gate (CG) Amplifier

This configuration works as a "current buffer": unity current gain.



$$v_i = -v_{gs}$$

$$g_m v_{gs} = -\frac{v_o}{R_D \parallel R_L} \qquad g_m v_i = \frac{v_o}{R_D \parallel R_L}$$

$$A_v = \frac{v_o}{v_i} = g_m (R_D \parallel R_L)$$

Input Resistance

$$R_i = \frac{1}{g_m}$$

Output Resistance
short-circuit source v_{sig}

$$R_o = R_D$$

Current gain

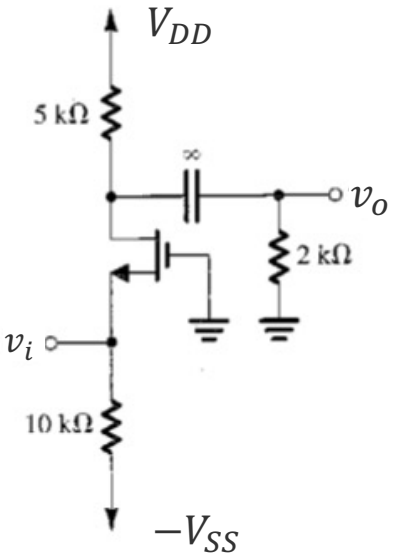
$$A_i = \frac{i_o}{i_i} = A_v \cdot \frac{R_i}{R_L}$$

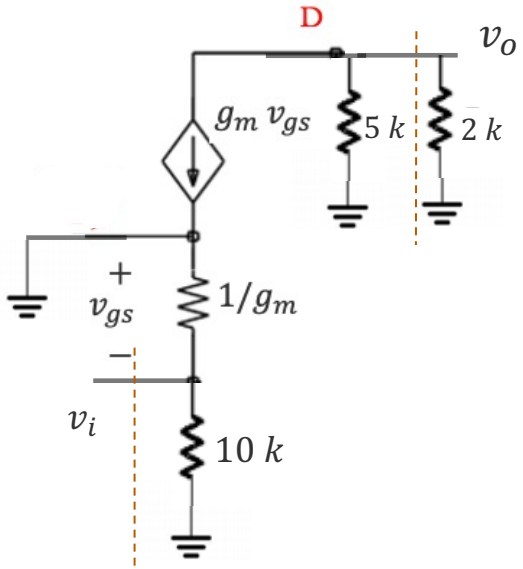
$$A_i = g_m (R_D \parallel R_L) \cdot \frac{1/g_m}{R_L}$$

$$A_i = \frac{(R_D \parallel R_L)}{R_L} \leq 1$$

Example

Analyze the amplifier given below. Given g_m=5 mA/V and r_o very high.





$$v_i = -v_{gs}$$

$$v_o = -g_m v_{gs}\,(5\ k \parallel 2\ k)$$

$$v_o = g_m v_i\,(5\ k \parallel 2\ k)$$

$$A_v = \frac{v_o}{v_i} = g_m(5\ k \parallel 2\ k)$$

$$A_v = 7.14\ V/V$$

Input Resistance

$$R_i = (10\ k \parallel 1/g_m)$$

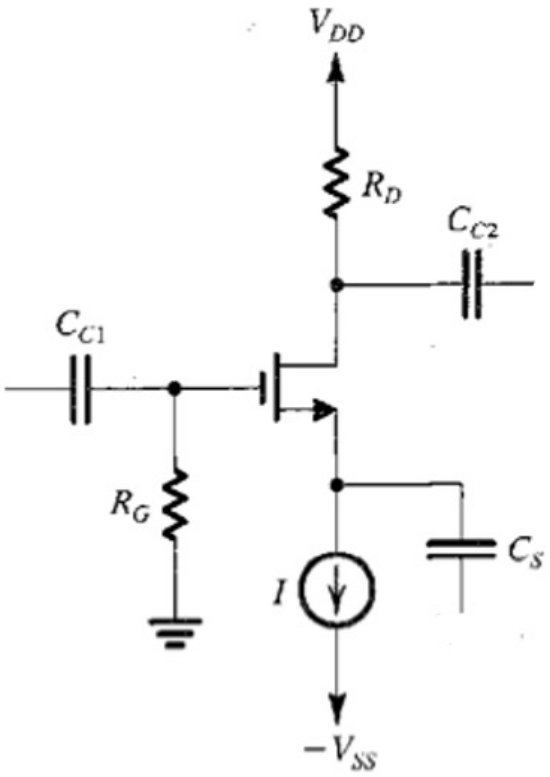
$$R_i = 196.1\ \Omega$$

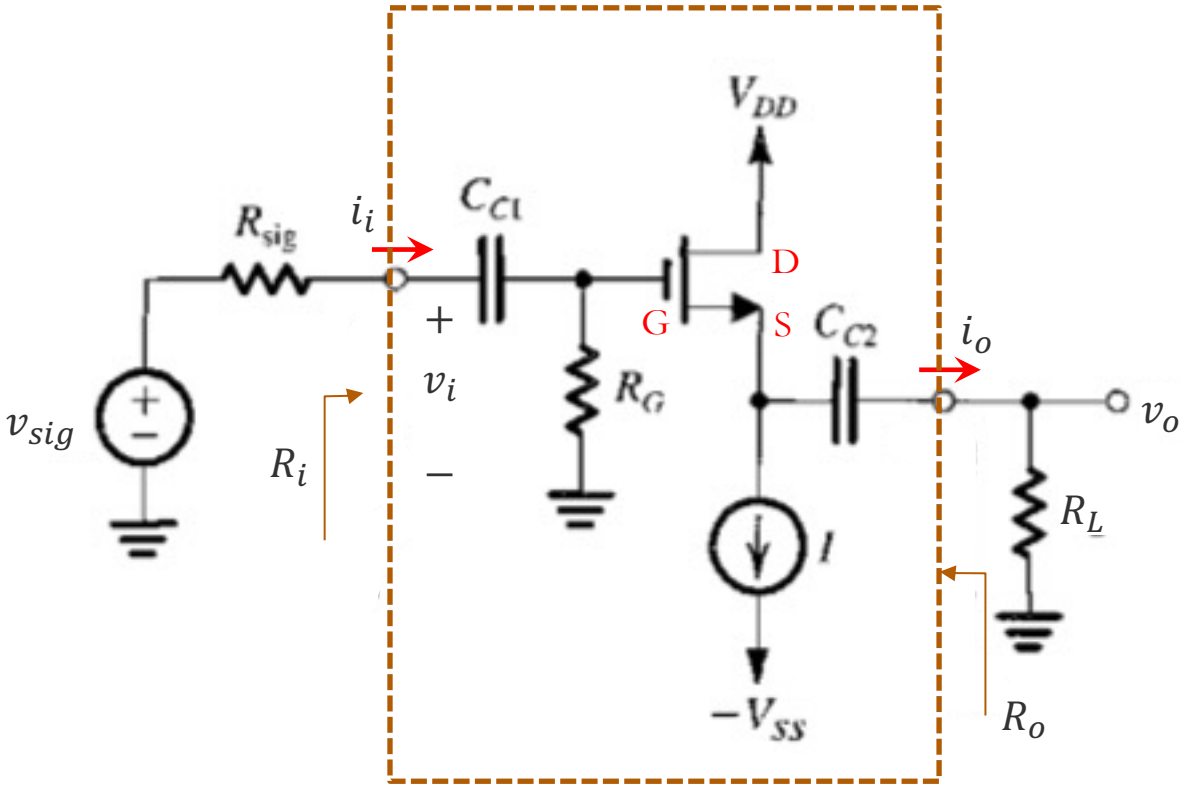
Output Resistance
short-circuit source v_i

$$R_o = 5\ k\Omega$$

The Common-drain (CD) Amplifier

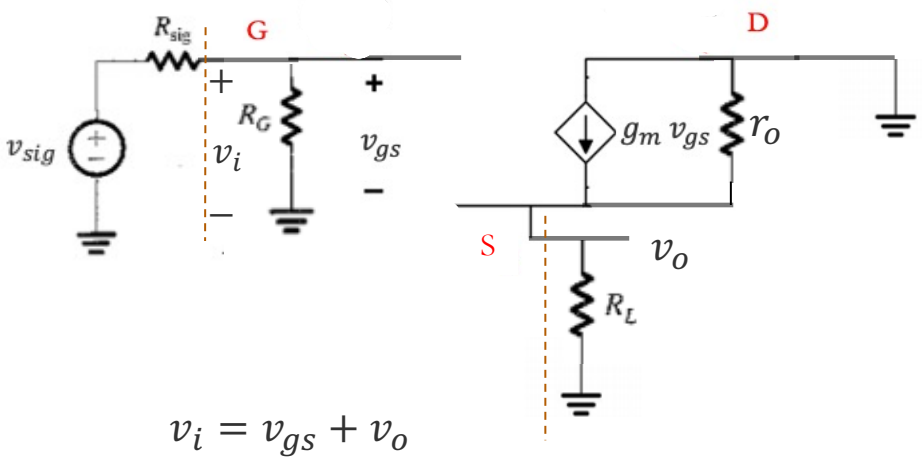
This configuration works as a "source-follower": unity voltage gain.





The Common-drain (CD) Amplifier

This configuration works as a "source-follower": unity voltage gain.



$$v_i = v_{gs} + v_o$$

$$g_m v_{gs} + \frac{0 - v_o}{r_o} = \frac{v_o}{R_L}$$

$$g_m(v_i - v_o) + \frac{0 - v_o}{r_o} = \frac{v_o}{R_L}$$

$$A_v = \frac{v_o}{v_i} = \frac{g_m}{g_m + 1/R_L + 1/r_o} \leq 1$$

Input Resistance

$$R_i = R_G$$

Output Resistance
short-circuit source v_{sig} $\quad R_o = \frac{v_o}{i_o}$

$$i_o = g_m(-v_o) + \frac{0 - v_o}{r_o}$$

$$R_o = \frac{v_o}{i_o} = \frac{1}{g_m + 1/r_o} \approx \frac{1}{g_m}$$

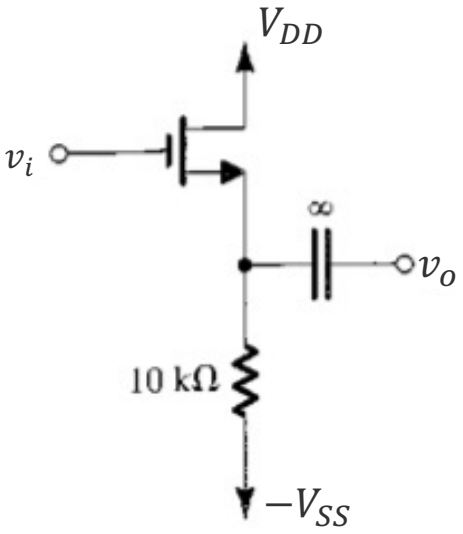
Current gain

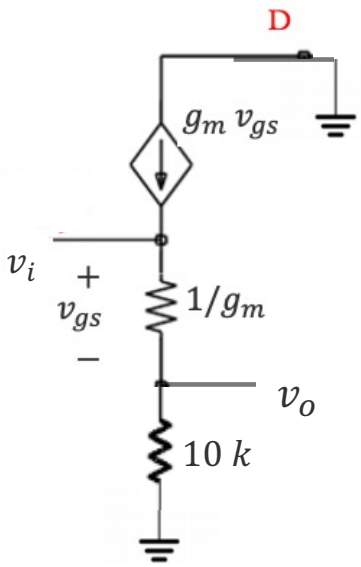
$$A_i = \frac{i_o}{i_i} = A_v \cdot \frac{R_i}{R_L}$$

$$A_i = \frac{g_m}{g_m + 1/R_L + 1/r_o} \frac{R_G}{R_L}$$

Examples

1) Analyze the amplifier given below. Given g_m=5 mA/V and r_o very high.





$$v_i = v_{gs} + v_o$$

$$v_o = g_m v_{gs}\,(10\ k)$$

$$v_o = g_m(v_i - v_o)(10\ k)$$

$$A_v = \frac{v_o}{v_i} = \frac{g_m\,(10\ k)}{1 + g_m\,(10\ k)}$$

$$A_v = 0.98\ V/V$$

Input Resistance

$$R_i = \infty$$

Output Resistance
short-circuit source v_i

$$R_o = \frac{v_o}{i_o}$$

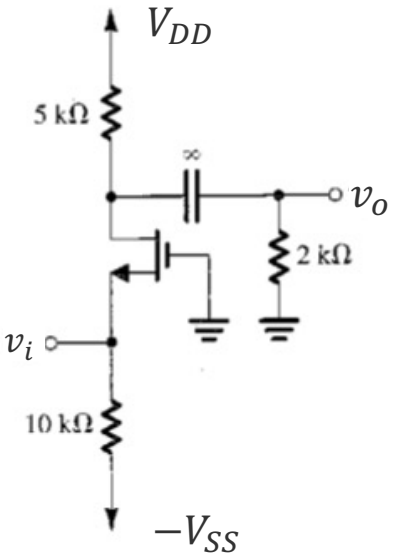
$$i_o = g_m v_{gs} - \frac{v_o}{10\ k}$$

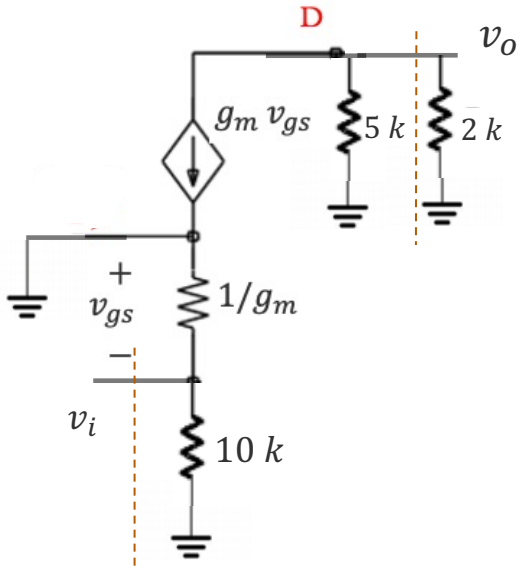
$$i_o = -g_m v_o - \frac{v_o}{10\ k}$$

$$R_o = \frac{1}{g_m + \frac{1}{10k}} = 196.1\ \Omega$$

Examples

2) Analyze the amplifier given below. Given g_m=5 mA/V and r_o very high.





$$v_i = -v_{gs}$$

$$v_o = -g_m v_{gs}\,(5\ k \parallel 2\ k)$$

$$v_o = g_m v_i\,(5\ k \parallel 2\ k)$$

$$A_v = \frac{v_o}{v_i} = g_m(5\ k \parallel 2\ k)$$

$$A_v = 7.14\ V/V$$

Input Resistance

$$R_i = (10\ k \parallel 1/g_m)$$

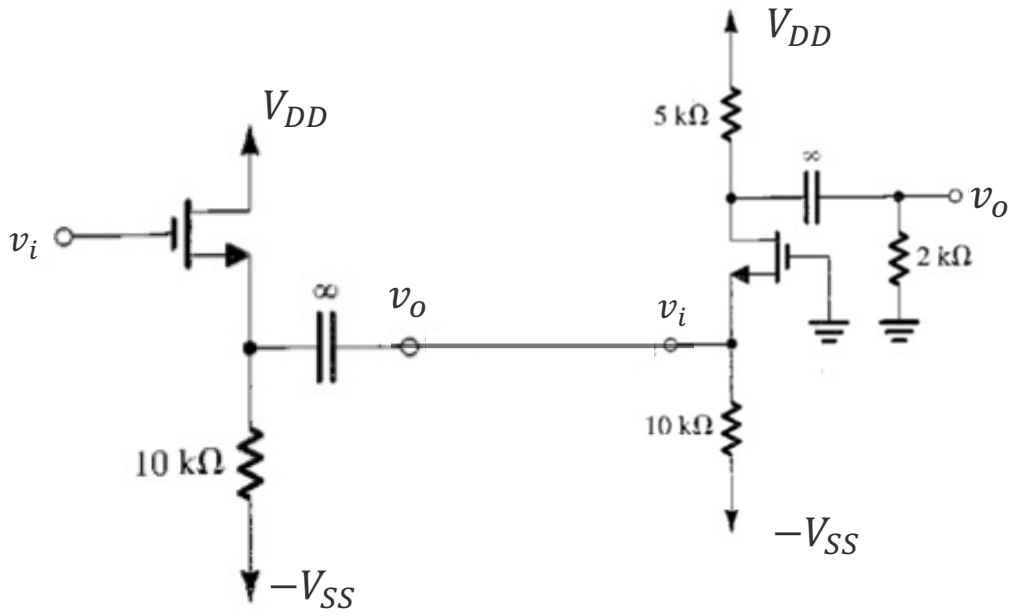
$$R_i = 196.1\ \Omega$$

Output Resistance
short-circuit source v_i

$$R_o = 5\ k\Omega$$

Examples

3) Connect the two circuits.



Multi-stage Amplifier Design

Voltage gain

$$A_v = \left(\frac{v_o}{v_i}\right)_1 \cdot \left(\frac{v_o}{v_i}\right)_2$$

$$A_v = 7\ V/V$$

Input Resistance

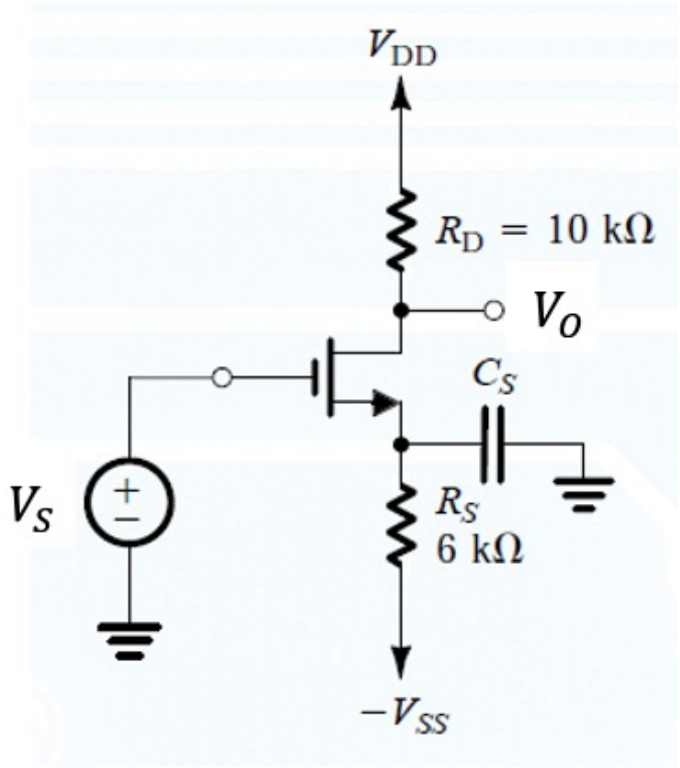
$$R_i = \infty$$

Output Resistance

$$R_o = 5\ k\Omega$$

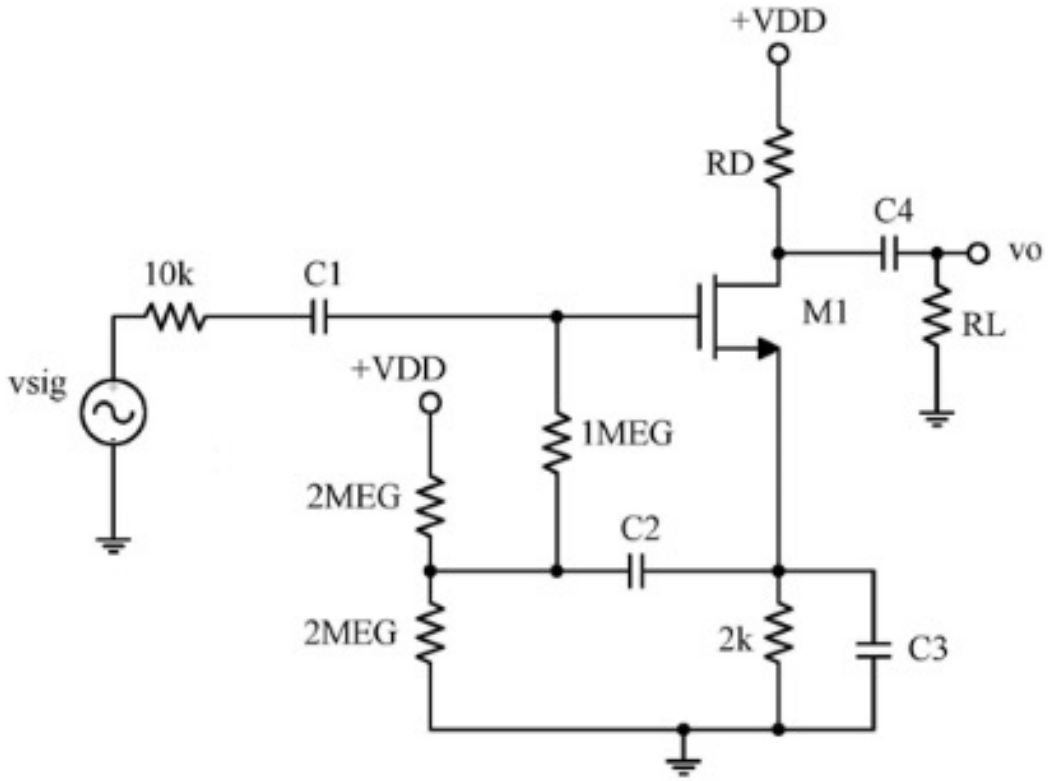
Examples

(4) Analyze the amplifier given below.



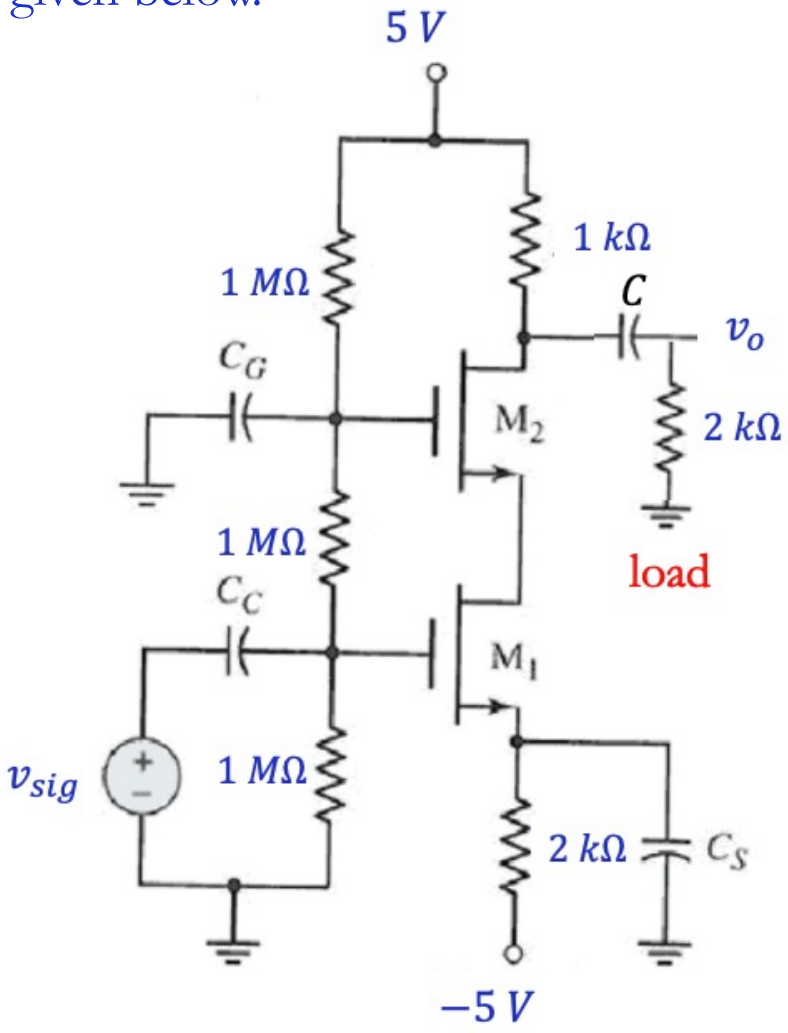
Examples

(5) Analyze the amplifier given below.



Examples

(6) Analyze the amplifier given below.

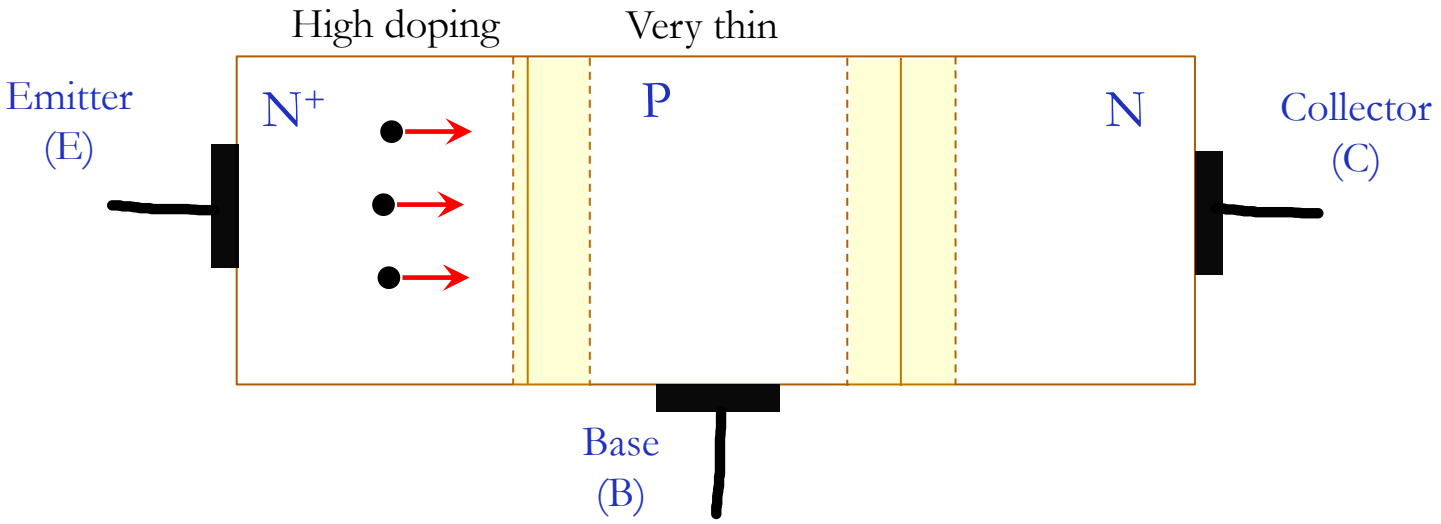


Bipolar Junction Transistors

- BJT Basic Structure.
- Current in BJTs and Operation Modes.
- DC Analysis (Biasing).
- Small Signal Analysis.
- The BJT as an Amplifier.

Device Structure

The bipolar junction transistor is basically two diodes (pn junctions) connected back-to-back.

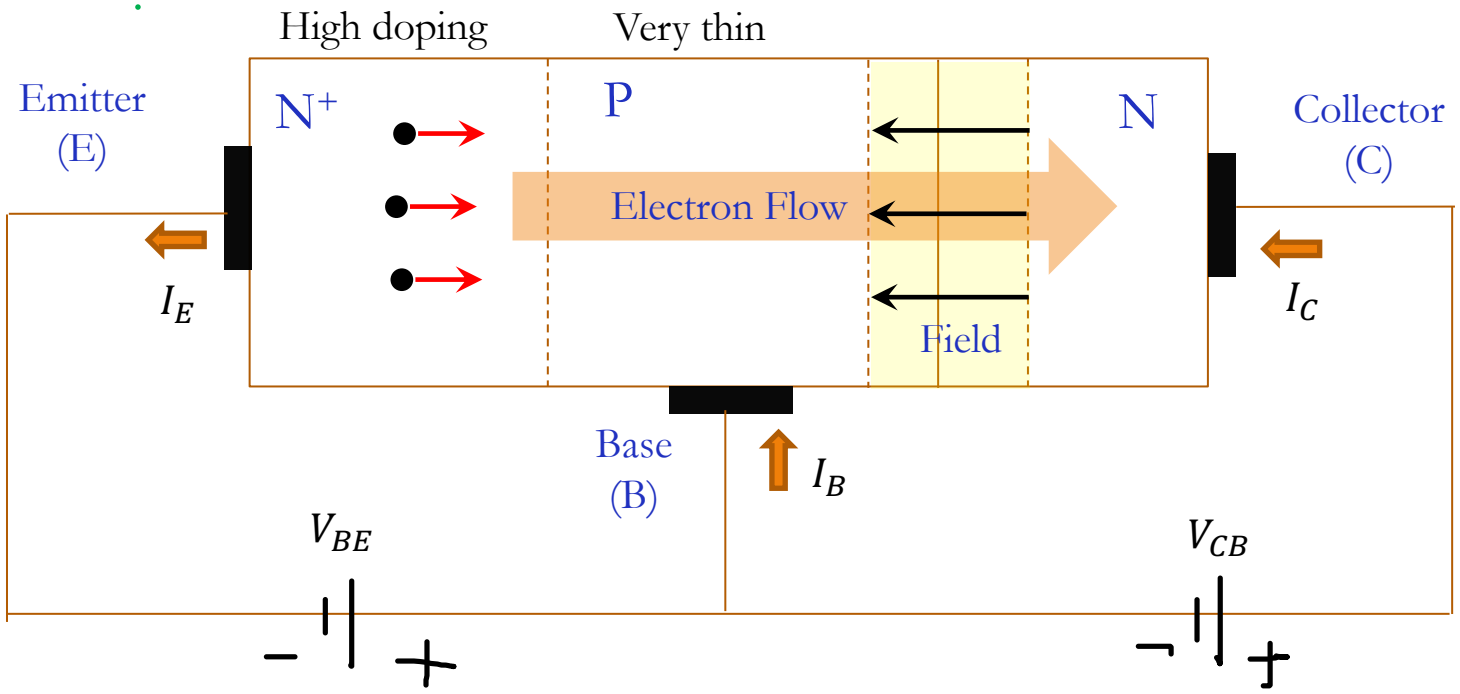


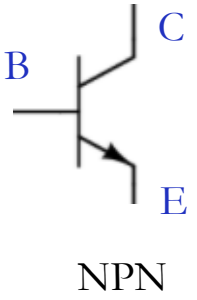
Four possible scenarios for current.

1) The two junction are reversed biased (cutoff mode).
2) BE junction forward, BC junction reversed (active mode).
3) BE junction reversed, BC junction forward (reverse active mode).
4) The two junction are forward biased (saturation mode).

The Active Mode

The active mode is the mode of operation we use in amplifier circuit.





Current relations

$V_{BE} = 0.7\ V$

$I_E = I_B + I_C$

$I_E = (\beta + 1) I_B$

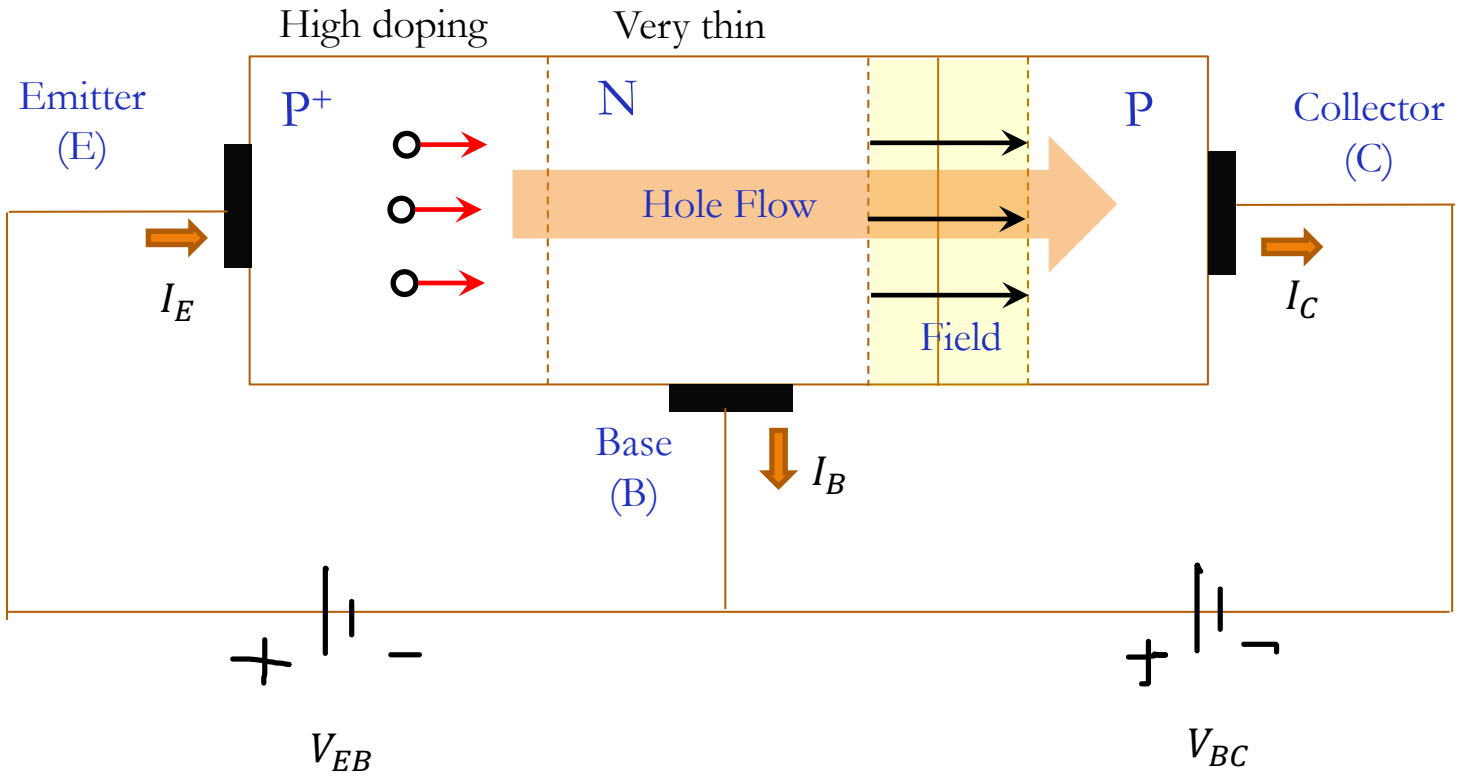
$\beta = \frac{I_C}{I_B}$

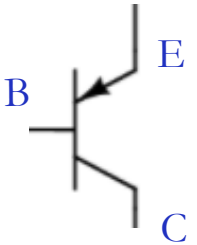
$\alpha = \frac{I_C}{I_E}$

$\alpha = \frac{\beta}{\beta + 1}$

The Active Mode

The bipolar junction transistor is basically two diodes (pn junctions) connected back-to-back.





PNP

Current relations

$$V_{EB} = 0.7\ V$$

$$I_E = I_B + I_C$$

$$I_E = (\beta + 1) I_B$$

$$\beta = \frac{I_C}{I_B}$$

$$\alpha = \frac{I_C}{I_E}$$

$$\alpha = \frac{\beta}{\beta + 1}$$

Q & A

Send me an email if
you have any
questions.